An explicit logarithmic coloring of the associahedron through heavy-path tree statistics

Benjamin Aram Berendsohn, Jean Cardinal, John Iacono, Laszlo Kozma

arxiv:2610.11426v1; CC BY 4.0

Editorial note. Original author coloring proof and both full TikZ figures retained. Original acknowledgement of AI-generated figures remains visible; no independent certification of authorship or correctness is asserted. One theorem is counted, not individual rotation cases.

Editorial note. Mechanical source0.tex adapter expands the two original TikZ input files in place; no mathematical or drawing content is changed. Original inputs and raw mainArxiv.tex retained. Bibliography math-case protection is limited to unescaped math expressions.

Editorial note. Mechanical natbib adapter braces the original MR3383157 year range [2014–2016] so its closing bracket cannot terminate the generated bibitem optional label. This preserves the original author-supplied entry, year range and cite key; no source is added or reconstructed.

Allowed standard prerequisites (editorial).

- Associahedra, binary-tree rotations and heavy-path decompositions. Original graph/tree equivalence, subtree sizes, heavy children, path-depth parity and exact color formula retained.. The nonroutine combined coloring construction and every rotation case remain in the counted proof.

Abstract

Addario-Berry, Reed, Scott, and Wood (JoCG 2026) proved that the n-dimensional associahedron has chromatic number at most $22500 \cdot \log_3 n + O(1)$. We present a much simpler proof that the chromatic number of the $n-1$-dimensional associahedron is at most $6 \log_2 n + 18$. Since this paper's appearance as a preliminary abstract in the Japan Conference on Discrete and Computational Geometry, Graphs, and Games held from September 7-10, 2026, the bound has been improved to $O(\log \log n)$ by Oum and Wood (arXiv September 30, 2026). It is hoped that the simple method here could give insights to further improvements.

Introduction. Associahedra are polytopes that appear in many areas of mathematics. We refer to the survey of Pilaud, Santos, and Ziegler Pilaud et al. [2023] for an overview of the developments around these polytopes in the past few decades. We are interested in the graphs of associahedra. For a given positive integer n, the vertices of the $(n-1)$-dimensional associahedron are in bijection with triangulations of a convex $(n+2)$-gon, or, via planar duality, with rooted binary trees on n (internal) nodes. Hence for a given n, the number of vertices is $C_n = \frac{(2n)!}{(n+1)!n!}$, the nth Catalan number. Edges of associahedra connect pairs of triangulations that differ by a single *diagonal-flip*, an operation that consists of replacing a diagonal of a quadrilateral formed by two triangles sharing an edge by the other diagonal. Equivalently — and this is the representation that we will use in our proofs — edges connect pairs of vertices corresponding to two binary trees that differ by a single rotation.

While being well-studied, the structure of the associahedron graph still remains mysterious. It is known to have diameter $2n-6$ for sufficiently large n Sleator et al. [2022], Pournin [2014], and to be Hamiltonian (see for instance Manneville and Pilaud [[2014–2016]] for a generalization). Chromatic properties of related graphs are also investigated Baralić et al. [2024], Gargantini et al. [2026], Molnar et al. [2025].

The natural question of the chromatic number of associahedra was posed by Fabila-Monroy, Flores-Peñaloza, Huemer, Hurtado, Wood, and Urrutia Fabila-Monroy et al. [2009] who showed a lower bound

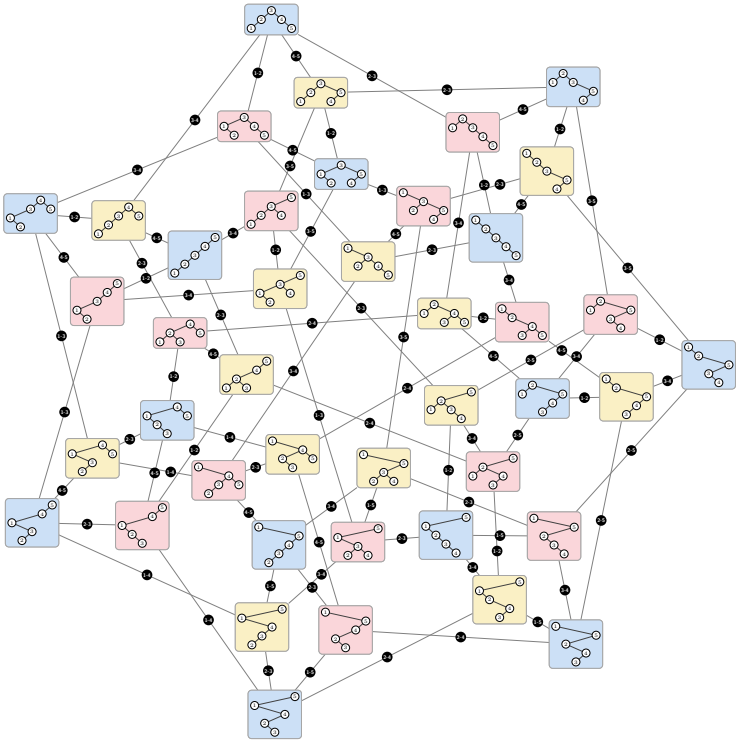


Figure 1: The graph of the 4-dimensional associahedron: Nodes are all binary search trees on $\{1, 2, 3, 4, 5\}$. Edges connect those nodes representing trees that are a single rotation apart where the nodes of the rotation appear in the black circle on each edge. The graph is 3-colored.

of 4 and an upper bound of $O(n/logn)$. It is only recently that Addario-Berry, Reed, Scott, and Wood Addario-Berry et al. [2026] showed that the chromatic number of associahedra is at most $22500 \cdot \log_3 n + O(1) = O(\log n)$. Their proof is elementary but quite involved. We give a simple proof that the chromatic number of the $(n-1)$-dimensional associahedron is at most $6 \log_2 n + 18$. This has since been improved to $O(\log \log n)$ Oum and Wood [2026] by Oum and Wood but we hope that the simple method here could give insights to further improvements.

Preliminaries. We make the following definitions, which lead up to assigning any binary tree T with n nodes an integer color in the range 0 to $6\lceil \log_2 n \rceil + 17$.

- If x is a node in some tree T, let $s_T(x)$ be the number of nodes in the subtree of x in T
- A node is *left heavy* if its left subtree is larger than its right subtree; otherwise it is *right heavy*.
- Let $L(T)$ be the set of left-heavy nodes in T.
- Let the *left color* of T, $c_\ell(T)$ be $|L(T)| \bmod 3$.
- The *heavy child* of a nonleaf node is its left child if it is left heavy and its right child if it is right heavy.
- Let $H(T)$ be the set of heavy children in T.
- Let the *heavy child color* of T, $c_h(T)$, be $\left\lceil \sum_{x \in H(T)} \log_2 s(x) \right\rceil \bmod \lceil \log_2 n + 1 \rceil$
- Let $P(T)$ be the set of *heavy* paths, which is the graph obtained by connecting each non-leaf node to its heavy child. The *heavy path depth* of a node is is its depth relative to the shallowest node (in the tree) on its heavy path. The *path color*, $c_p(T)$, is the parity of the number of left heavy nodes of even path depth.
- Finally, let the *color* of T, $c(T)$ be $6c_h(T) + 2c_\ell(T) + c_p(T)$. This is an integer in the range 0 to $6\lceil \log_2 n + 1 \rceil + 4 + 1 \leq 6 \log_2 n + 17$. It is designed so that if either the heavy child color or the left color or the path color differs between two trees then the color differs.

We now state the theorem:

Theorem 1. *The chromatic number of the* $(n-1)$*-dimensional associahedron is at most* $6 \cdot \log_2 n + 18$.

Proof. Given the above scheme, all that remains to prove is that a single rotation always changes the color of the tree.

Consider a rotation that transforms a tree T to T'. We assume the rotation is on two nodes x and y where y is the right child of x in T and x becomes the left child of y in T'; the reverse case is symmetric. We consider 16 cases based on whether x and y are each left or right heavy before and after the rotation. These are illustrated in Figure 2. These are split into four groups, those where the left color changes, those where the heavy child color changes, and finally those cases that are impossible.

Cases II-V, VIII, IX, XII-XV (Left color changes)

In these cases the number of left heavy nodes, $|L(T)|$ changes by 1 or 2, and thus $c_\ell(T)$ which is $|L(T)|$ mod 3 changes.

Cases I and XVI (Heavy child color changes)

For I (XVI is symmetric): Here the increase in $\sum_{x \in H} \log_2 s(x)$ is $\log_2(|A| + |B| + 1) - \log_2 |B| = \log_2 \frac{|A|+|B|+1}{|B|}$. As $|A| > |B|$, $\log \frac{|A|+|B|+1}{|B|} \geq \log_2 \frac{|B|+|B|+1}{|B|} > 1$. Also, $\log_2 \frac{|A|+|B|+1}{|B|} \leq \log_2 n$. Thus $\sum_{x \in H} \log_2 s(x)$ changes by at least 1 and at most $\log_2 n$, which causes $c_h(T) = \lceil \sum_{x \in H} \log_2 s(x) \rceil$ mod $\lceil \log_2 n + 1 \rceil$ to change.

Case X

The key point of this case is that the heavy paths do not change, with the one exception that x and y, which are adjacent on the same heavy path, have their positions change; x remains right heavy and y remains left heavy. Either x before (right heavy) and y after (left heavy) are at even depth on their path or y before (left heavy) and x after (right heavy) are at even depth; in either case the number of left heavy nodes of even path depth changes by one and thus the path color changes.

Cases VI, VII and XI (Impossible)

These are impossible. For VI (XI is symmetric) the following contradictions occur:

- $|A| > |B| + |C| + 1 > |B|$ and
- $|A| \leq |B|$

For VII:

- $|A| > |B| + |C| + 1 > |C|$ and
- $|C| \geq |A| + |B| + 1 > |A|$

Acknowledgments Jean Cardinal and John Iacono's research is supported by the F.R.S.–FNRS. Generative AI was used to create the Tikz figures but did not write a single word and was not used to obtain the result.

References

Louigi Addario-Berry, Bruce Reed, Alex Scott, and David R. Wood. A logarithmic bound for the chromatic number of the associahedron. *Journal of Computational Geometry*, 17(1):61–74, May 2026. doi: 10.20382/jocg.v17i1a3. URL `https://jocg.org/index.php/jocg/article/view/5192`.

Dj. Baralić, J. Ivanović, and Z. Petrić. Chromatic numbers for facet colouring of some generalised associahedra. *Discrete Appl. Math.*, 358:272–284, 2024. ISSN 0166-218X,1872-6771. doi: 10.1016/j.dam.2024.07.013. URL `https://doi.org/10.1016/j.dam.2024.07.013`.

Ruy Fabila-Monroy, David Flores-Peñaloza, Clemens Huemer, Ferran Hurtado, David R. Wood, and Jorge Urrutia. On the chromatic number of some flip graphs. *Discret. Math. Theor. Comput. Sci.*, 11 (2):47–56, 2009. doi: 10.46298/DMTCS.460. URL `https://doi.org/10.46298/dmtcs.460`.

Ana Gargantini, Adri√°n Pastine, and Pablo Torres. Effect of graph operations on the structure and chromatic number of rotation graphs. *Annals of Combinatorics*, 2026. ISSN 0219-3094. doi: 10.1007/s00026-026-00821-1.

Thibault Manneville and Vincent Pilaud. Graph properties of graph associahedra. *Sém. Lothar. Combin.*, 73:Art. B73d, 31, [2014–2016]. ISSN 1286-4889.

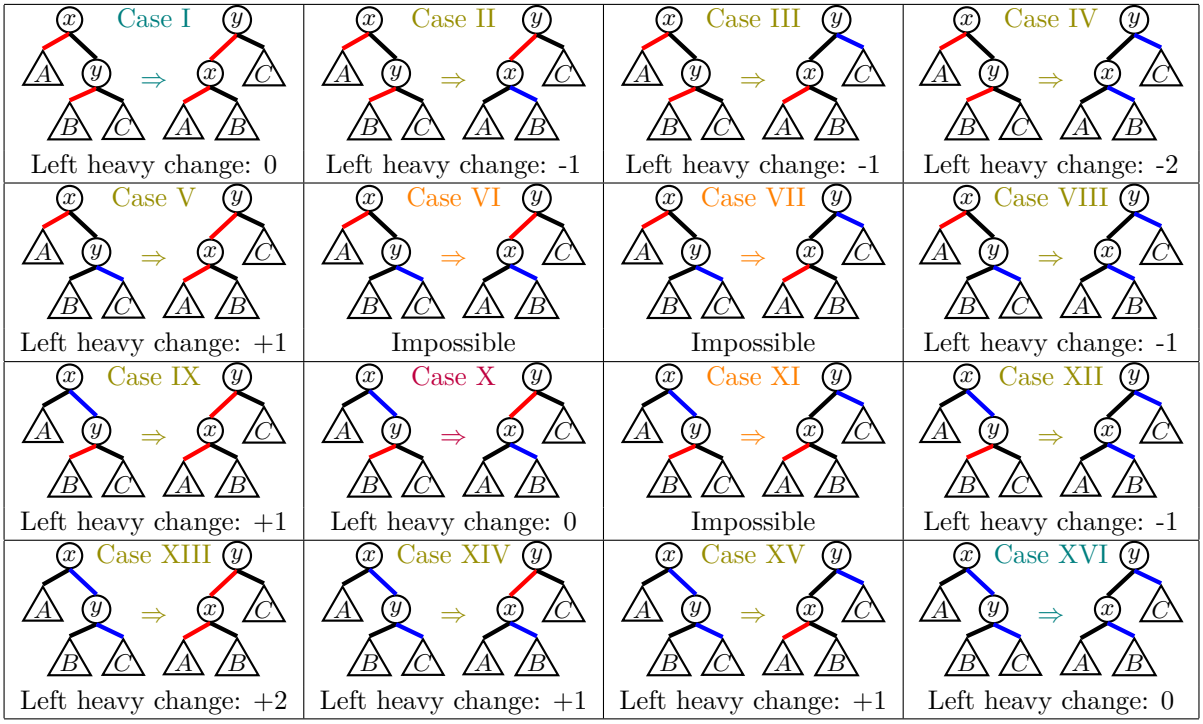


Figure 2: The cases. We consider a left rotation on x with right child y. A red edge indicates a left heavy node above and a blue edge indicates a right heavy node above. We consider all 16 possible cases of left and right heaviness for x and y before and after the rotation. In Cases II-V, VIII, IX, XII-XV the number of left heavy nodes change by 1 or 2 and thus the left color changes. In Case X the only change in heavy paths is the relative ordering of x and y is swapped, each retaining their heaviness; thus the path color changes. Cases VI, VII and XI are impossible. Cases I and XVI are possible but have no change in the number of left heavy nodes and thus require a more detailed argument and are the source of the logarithmic number of colors.

Anton Molnar, Cosmin Pohoata, Michael Zheng, and Daniel G. Zhu. A Lovász-Kneser theorem for triangulations, 2025. URL https://arxiv.org/abs/2510.27689. arXiv preprint 2510.27689.

Sang-il Oum and David R. Wood. A double-logarithmic upper bound on the chromatic number of the associahedron, 2026. URL https://arxiv.org/abs/2609.39015. arXiv preprint 2609.39015.

Vincent Pilaud, Francisco Santos, and Günter M. Ziegler. Celebrating Loday's associahedron. *Arch. Math. (Basel)*, 121(5-6):559–601, 2023. ISSN 0003-889X,1420-8938. doi: 10.1007/s00013-023-01895-6. URL https://doi.org/10.1007/s00013-023-01895-6.

Lionel Pournin. The diameter of associahedra. *Adv. Math.*, 259:13–42, 2014. ISSN 0001-8708,1090-2082. doi: 10.1016/j.aim.2014.02.035. URL https://doi.org/10.1016/j.aim.2014.02.035.

Daniel D. Sleator, Robert E. Tarjan, and William P. Thurston. Rotation distance, triangulations, and hyperbolic geometry. In *Collected works of William P. Thurston with commentary. Vol. III*, pages 343–377. Amer. Math. Soc., Providence, RI, 2022. ISBN 978-1-4704-6390-8; [9781470468354]; [9781470451646].